Kirill Kozlov

Software Engineer (Backend, Distributed Services, Applied AI)

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Summary

Software engineer and Machine Learning master's student at CMC MSU with production experience at X5 Tech. Built Python services using FastAPI, PostgreSQL, and Kubernetes; developing AI product search and catalog-processing pipelines for Korzina.

Technical Skills

Programming languages: Python, C++, Go, SQL, TypeScript
Fundamentals: data structures and algorithms, system design, REST API design, asynchronous programming, relational data modeling, testing (pytest), code review
Backend and infrastructure: FastAPI, SQLAlchemy (async), Alembic, Pydantic, asyncio, httpx, PostgreSQL, Redis, Apache Kafka, Docker, Kubernetes (K8s), Helm, GitLab CI/CD, Grafana, Linux, Git
Applied AI: Large Language Models (LLM), agentic workflows, RAG, prompt engineering, LangChain, OpenAI API, OCR (Optical Character Recognition), A/B testing, PyTorch
Frontend: React, TypeScript
Spoken languages: Russian — Native, English — B2

Work Experience

Software Engineer (Junior) | X5 Tech (X5 Group, Russia's largest grocery retailer), Logistics Solutions Department, Moscow | October 2026 — Present

- Developing the system that handles 90% of all foreign supplies X5 Retail Group purchases for its retail chains.

Software Engineer (Intern) | X5 Tech, AI Agents Department, Moscow | April 2026 — October 2026

- Cut legal opinion preparation time from about 97 minutes to 8–19 minutes (8 FTE saved for X5 Group) by building an agentic LLM contract-review system end to end: FastAPI and PostgreSQL backend, React and TypeScript frontend, multi-agent LangChain pipeline (document processing, requisite extraction, counterparty and property checks, risk analysis, legal opinion) with contract sections analyzed in parallel; over 500 reviews in production in May–August 2026.
- Kept minute-long OCR and LLM calls reliable in production by adding per-request budgets and timeouts, per-agent latency metrics in Grafana, and Kubernetes dev and prod environments deployed via Helm and GitLab CI/CD.
- Grounded legal analysis in Russian legislation by building a RAG source over 40 federal laws (366 articles, BM25 index) with relevant articles injected into agent prompts.
- Shipped an OCR and LLM service that reads foreign invoices for X5 Tech: from first commit in June 2026 to dev and prod deployment and a pilot in August 2026.
- Saved 10 FTE on utility-bill processing across X5 Group premises by developing the FastAPI backend that consumes OCR results from Apache Kafka, stores them and serves them to the frontend.

Projects

Korzina — Marketplace and AI Product Search | TypeScript, Node.js, PostgreSQL, React | September 2026 — Present

- Building a Telegram Mini App and bot with a catalog of 100,000+ products; automated supplier-data ingestion, normalization, and deduplication.
- Implemented PostgreSQL full-text and trigram search; developing a hybrid retrieval prototype combining multimodal Gemini embeddings, pgvector, reciprocal rank fusion (RRF), and Qwen reranking.
- Implemented LLM/VLM attribute extraction and product-compatibility checks in the prototype, with versioned embedding caches, concurrent-request deduplication, and incremental refresh.
- Built offline relevance evaluation with frozen datasets, qrels, Recall@20, and Precision@5; compared text and multimodal retrieval under model-cost and latency constraints.
- Built the Fastify API and background jobs; deployed the service using PostgreSQL, Valkey, S3, Docker, and GitLab CI/CD.

Other projects: AxiomAI: LLM-powered Telegram bot, 80% reduction in manager workload; WB Parser: Wildberries parser and Mini App, 220 users; Placa: analytics for 300 sellers.

Education

Lomonosov Moscow State University, Master's degree in Machine Learning | Faculty of Computational Mathematics and Cybernetics (CMC), Dept. of Mathematical Methods of Forecasting | September 2026 — Present

Lomonosov Moscow State University, Bachelor's degree in Computer Science | Faculty of Computational Mathematics and Cybernetics (CMC) | August 2022 — August 2026

- Deep Learning School (DLS), Computer Vision and NLP tracks — MIPT School of Deep Learning.
- Summer School on A/B Testing, X5 Tech — Summer 2026 (certificate no. 9074503).